

Toward a Resolution of the Hodge Conjecture via Algebraic Convergence

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Abstract

We propose a strategy for resolving the Hodge Conjecture by constructing rational Hodge classes as limits of sequences of algebraic cycle classes. This framework incorporates tools from sheaf cohomology, derived categories, and algebraic topology, and reframes transcendental obstructions as convergent anomalies in MayerVietoris constructions. We outline a pathway from approximable cycles to canonical representatives through a derived-categorical and motivic lifting process.

1 Problem Statement

Let X be a smooth projective complex variety. The classical Hodge Conjecture asserts:

Every class $\gamma \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ is the image of an algebraic cycle class:

$$\exists Z \in Z^p(X) : \quad \text{cl}(Z) = \gamma.$$

In known cases, obstructions to constructing such Z often appear in higher cohomology groups, particularly as connecting morphisms in long exact sequences:

$$\delta(\alpha_U - \alpha_V) = \tau \in H^{2p+1}(X).$$

2 Core Proposal

We replace the direct construction with a convergence framework:

1. **Construct a sequence of algebraic cycles** $Z_n \in Z^p(X)$ such that

$$\lim_{n \rightarrow \infty} \text{cl}(Z_n) = \gamma_\infty \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X).$$

2. **Interpret gluing mismatches** $\tau_n = \delta(\alpha_U^{(n)} - \alpha_V^{(n)}) \in H^{2p+1}(X)$ as a converging error term:

$$\lim_{n \rightarrow \infty} \tau_n = \tau.$$

3. **Lift the obstruction** τ to a trivial class in a derived category or via a motivic extension, such that it becomes exact or cancelable.
4. **Compactify the sequence** $\{Z_n\}$ into a single algebraic cycle Z through a convergence-preserving functor from a moduli space, stack, or formal scheme.

3 Strategic Hypothesis

We hypothesize:

Algebraic Approximability of Hodge Classes: Every rational Hodge class is the limit of a sequence of algebraic cycle classes.

4 Technical Goals

- Define and prove convergence of $\{Z_n\}$ under a Hodge-norm topology.
- Construct a categorical compactification functor $\widehat{\text{cl}}: \widehat{Z}^p(X) \rightarrow H^{2p}(X, \mathbb{Q})$.
- Show that the limiting class γ_∞ corresponds to an actual cycle via patching, degeneration, or formal limit stabilization.
- Show that all $\tau \in H^{2p+1}(X)$ arising from such sequences are killed in a derived or motivic category.

5 Conclusion

If successful, this program will convert a transcendental obstruction into a geometric convergence process, yielding a genuine algebraic cycle representative for every rational Hodge class and completing the classical conjecture.